

MD SHAH IMRAN SHOYON

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• Graduate Research Assistant, University of Florida • shah-imran.github.io • Gainesville, Florida

EDUCATION

Ph.D. in Electrical & Computer Engineering (Ongoing)

January 2025 – Expected July 2029

University of Florida, Gainesville, Florida

CGPA: 3.73/4.00

Relevant Coursework: Image Processing and Computer Vision, Fundamental Machine Learning, Semiconductor Packaging, Physical Assurance and Inspection of Electronics, Reconfigurable Computing, Computational Photography

B.Sc. in Electrical & Electronic Engineering

July 2020

Shahjalal University of Science and Technology, Sylhet, Bangladesh

Relevant Coursework: Signals & Linear Systems, Digital Signal Processing, Numerical Analysis

PROFESSIONAL EXPERIENCE

PhD in ECE (Ongoing), University of Florida, GNV, FL, USA

Jan 2025 – Present

Graduate Research Assistant

- Conducted research across physical assurance, counterfeit IC detection, X-ray inspection, multimodal AI, and 3D medical image segmentation, including dataset development, model training, and inspection workflows.
- Researching agentic security, with a focus on identifying and evaluating agentic-like attack surfaces, multi-stage attack cascades, and security risks in LLM-based systems.
- Developing neurosymbolic AI (NeSy) approaches for hardware verification of Systems-on-Chip (SoCs), combining LLM-based reasoning with symbolic constraints, verification history, and traceable validation workflows.

HowardMiller, Zeeland, USA

Oct 2021 – Dec 2024

Full Stack Developer (Remote)

- Developed AI-driven deep learning strategies to optimize PPC budget allocation, reducing spend by 20% while maintaining the same sales volume (TensorFlow, Keras, Pandas, Scikit-Learn).
- Designed and optimized features, integrating third-party APIs using Flask, Django, Celery, Redis, Postgres, AWS, React.js, and related tools.
- Collaborated with stakeholders to gather and document user requirements, translating them into technical specifications and developing optimized features.

Gmonster.co, Balatonalmádi, Hungary

Sep 2018 – Oct 2021

Project Lead (Remote)

- Oversaw the full development pipeline and built a Mass Email Marketing App with a subscription system from scratch.
- Engineered the system to send 10K+ emails per instance using separate proxy pipelines, leveraging concurrency and async processing for 20% higher efficiency than competitors.
- Achieved rapid adoption with 500+ users acquired within the first 2 months.

EnableGeek, Dhaka, Bangladesh

Jun 2022 – Dec 2024

Founder

- Designed SEO pipeline, growing traffic to 500+ daily clicks within a year.
- Managed content and tech teams; published two books on Python and Java.

PROJECTS HIGHLIGHTS

Neurosymbolic AI for SoC Hardware Verification | Research Project

2026 – Present

- Developing a NeSy verification framework that combines LLM reasoning with symbolic constraints and hardware verification knowledge.
- Integrating traceable verification history and structured evidence to improve grounding, explainability, and reuse of prior verification results.

- Exploring automated reasoning workflows for SoC security verification using LLM-assisted analysis with formal and symbolic validation.

Agentic Security for LLM-Based Systems | Research Project 2026 – Present

- Investigating agentic-like attacks that exploit fabricated or manipulated interaction surfaces to influence LLM-driven decision making.
- Designing and evaluating multi-vector, multi-stage attack scenarios to study cascading security failures in agentic systems.
- Developing benchmark-style evaluation pipelines for measuring attack success, robustness, and failure propagation across system components.

Physical Inspection of Electronics: Counterfeit Detection | Research Project 2025 – Present

- Designed pinhole-image pipeline with automated defect ROI extraction for IC packages.
- Trained CNN/Transformer classifiers on in-house dataset for counterfeit detection with high precision/recall.
- Built multimodal optical-image dataset and VLM training setup for extended classification and explainability.

LLaMA Fine-Tuning with LoRA & QLoRA | Research Project 2025 – Present

- Contributed to physical-inspection dataset development (collection, labeling standards, QC).
- Set up reproducible training environments (Pipenv, Pyenv) on lab servers.
- Assisted in distributed training (torchrun) and optimized codebase (modularization, speedups, configs).
- Added pytest coverage and conducted code reviews for style and reliability.

Machine Learning | Artificial Intelligence | Computer Vision Projects 2017 – Present

- Analyzing the Consequences of High-Intensity Light YOLOv5 and Strategies for Defending Against Adversarial Attacks (Yolo, Python) [GitHub](#)
- Custom Object Detection and Tracking using Darknet (Darknet-yolo, Python) [GitHub](#), [Youtube](#)
- Dataset Creation for Training Custom Yolo Model (Yolo, Linux Shell Scripting) [Drive](#)

Academic Projects 2014 – 2020

Collaborated as a "Research Assistant" on three Official Projects supported by the UGC (University Grants Commission), leading the last two initiatives as Team Leader.

- Built an Autonomous Underwater Vehicle with navigation system (Arduino, Microcontroller).
- Developed an RFID-based automatic attendance system (Python, Sockets, SQL).
- Designed a UGV and Human Assistant system (Python, Raspberry Pi, Image Processing, LiDAR).

PUBLICATION

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- Al, Zaman, M., Jawad, A., Santamaria-Pang, A., Lee, H. H., Tarapov, I., See, K., **Imran, M. S.**, Roy, A., Fallah, Y. P., Asadizanjani, N., & Forghani, R. (2025). **WaveFormer: A 3D Transformer with Wavelet-Driven Feature Representation for Efficient Medical Image Segmentation**. ArXiv.org. <https://arxiv.org/abs/2503.23764> (MICCAI 2025)
 - **Shah, M.**, Shajib Ghosh, Istiaq Firoz Shiam, Patrick Craig, Antika Roy, Pratyush Shukla, Varun Sai Vemuri, Hafez Bahrami, and Navid Asadizanjani (2026) **GUIDE-X: VLM-LLM-based AI assistant for artifact detection and operator guidance in x-ray inspection** (Accepted 2026 SPIE)
 - Shajib Ghosh, Patrick Craig, Antika Roy, **Md Shah Imran Shovon**, Varun Sai Vemuri, John Allgair, Hafez Bahrami, and Navid Asadizanjani (2026) **AURA-PIE: an AI-enabled understanding and retrieval-augmented assistant for physical inspection of electronics** (Accepted 2026 SPIE)
 - **PINCLASS: Pinhole-Centric Counterfeit Classification of IC Packages**. **Md Shah Imran Shovon**, Chih-Yun Pai, Shajib Ghosh, Chien-Chia Huang, Patrick J. Craig, Navid Asadizanjani. (PAINE 2025).
 - **A Novel Framework for Identifying Counterfeit ICs via Pinhole Evaluation**. **Md Shah Imran Shovon**, Shajib Ghosh, Patrick J. Craig, Chien-Chia Huang, Chih-Yun Pai, Navid Asadizanjani. (RAPID 2025).
 - **Deep Learning-Driven X-ray Analysis for High-Throughput Manufacturing**. Patrick J. Craig, Antika Roy, Shajib Ghosh, **Md Shah Imran Shovon**, Nitin Varshney, Navid Asadizanjani. (RAPID 2025).

- **Shah, M. (2024). Evaluating YOLO Object Detection Models Under Adverse Lighting Conditions.** Bulletin of the American Physical Society; American Physical Society. <https://meetings.aps.org/Meeting/MAR24/Session/NN00.63>
- **Shah, M. (2024). Evaluating and Enhancing Image Caption Generation: A Comparative Study of LSTM and GRU Models with Object and Feature Recognition Strategies.** Bulletin of the American Physical Society; American Physical Society. <https://meetings.aps.org/Meeting/MAR24/Session/FF00.50>
- For a complete list of publications, please visit my Google Scholar profile: **Link**

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C++, C, VHDL, Matlab, Processing, Arduino, Flutter, Dart, Assembly Language (8086)

Libraries: PyTorch, TensorFlow, Keras, YOLO, NumPy, Scipy, Pandas, Selenium, Requests, BeautifulSoup4, Scrapy, Matplotlib, Seaborn, Bokeh, Django, Flask, React, Typescript,

Databases: SQL, Firebase, MongoDB, Pickle

Environments: Anaconda Environment, Jupyter Notebook, Pipenv

Hardware & Reconfigurable Computing: FPGA, VHDL

TRAINING & EXTRA CURRICULAR ACTIVITIES

- Led and participated in 12+ national robotics competitions (Line Follower, Maze/Grid Solver, Robo Fights) and electronics fairs, achieving 1st runner-up at Robomania, IUT (2015) with a multi-bot system (line following, obstacle avoidance, RC object carrier).
- Completed Top Up IT Java training by Ernst & Young (LICT project), certified by George Washington University.
- Completed AWS Cloud Practitioner course (2020), covering analytics, compute, serverless, containers, databases, and networking.